

Vatsal Omar

Software Engineer — AI Agents & Distributed Systems

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Software engineer with deep expertise in multi-agent AI systems, LLM orchestration, and full-stack product development. Built production agentic systems at BLive serving Zomato, TVS, Ather, and Bounce. Strong background in reinforcement learning (GRPO), multi-provider LLM integration, and real-time IoT data pipelines.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

2023-08 – 2026-07

Bachelor of Science in Computer Science — Pilani, IN

EXPERIENCE

Software Development Intern

2024-12 – 2025-09

BLive — Bangalore, IN

- Designed and shipped an agentic notification workflow using Firebase Cloud Messaging and a Python event-processing backend to route SOS alerts, delivery updates, and anomaly flags to the right operator roles with sub-second latency.
- Built a telematics intelligence layer consuming real-time IoT data streams from connected EVs, applying rule-based and ML-assisted anomaly detection to surface battery health degradation, SOS events, and route deviation alerts.
- Implemented Role-based Access Control (RBAC) system using NestJS + PostgreSQL with JWT Auth, enforcing scoped permissions for rider, fleet operator, and admin roles across a production system serving Zomato, TVS, Ather, and Bounce.
- Integrated Google Maps with live vehicle markers, geofencing, and dynamic route overlays on GCP-backed infrastructure, enabling real-time fleet visibility and route optimization for operations teams.
- Profiled and resolved rendering bottlenecks using Flutter DevTools, reducing app load times by ~25% across two production apps with high-frequency telemetry update cycles.
- Built two cross-platform Flutter apps (rider delivery + fleet dashboard) with shared Dart codebase across Android and Web; used Bloc and Riverpod for high-frequency state management.

PROJECTS

ChaosOps AI — Multi-Agent Incident Response Simulator

Python · FastAPI · TRL/GRPO · OpenAI · Anthropic · Docker

Hierarchical multi-agent system where SRE, Developer, Manager, and Oversight agents coordinate under partial observability to resolve cascading production failures. Includes rogue-agent detection, GRPO policy optimization, and deterministic benchmarking.

- Built LLM-agnostic agent interfaces with function-calling, streaming parsing, retry logic, and fallback handling across OpenAI, Anthropic Claude, and open-source providers.
- Designed rogue-agent detection where Oversight agent infers malicious behavior from telemetry and communication patterns, enabling detection of AI-induced failures.
- Fine-tuned agent policies using GRPO reinforcement learning with reward decomposition for resolution accuracy and oversight quality — 98 tests, CI-gated across Python 3.10/3.11/3.12.
- Implemented deterministic episode replay: (failure_type, seed) uniquely reproduces every trajectory, locked by a golden-trace integration test.
- Trained on Qwen 2.5 0.5B with Unsloth 4-bit LoRA: mean reward improved from -8.87 to +4.29 across 800 gradient steps with clean cross-zero around step 340.

StrategyVault — AI Trading Strategy Marketplace

FastAPI · Next.js 14 · Redis · PostgreSQL · Docker · Gemini 2.0

Subscription-based platform where AI generates, validates, and ranks institutional-grade trading strategies combining Gemini 2.0, Claude, and OpenAI in a multi-model consensus engine.

- Built multi-model Swarm consensus engine querying Gemini, OpenAI, and Claude in parallel to produce BUY/HOLD/REJECT verdicts with 0-100 confidence rankings.

- Engineered RBI (Research-Backtest-Implement) agent loop where Gemini 2.0 autonomously generates, debugs, and iterates backtesting code via structured prompt refinement.
- Integrated walk-forward validation, Monte Carlo simulation, and market regime detection to prevent overfitting; sandboxed execution with Redis caching and rate limiting.
- Shipped full-stack: FastAPI backend (249 tests) + Next.js 14 frontend with subscription tiers.

AcordLayer — Self-Hosted Collaboration Platform

Go · TypeScript · React · PostgreSQL · WebSockets · Docker

Self-hosted collaboration platform combining real-time chat, workflow automation, and AI-driven extensibility — built with Go and React for privacy-conscious teams.

- Architected workflow automation layer with no-code and API-based triggers for agentic task execution, containerized via Docker Compose for on-premises and cloud deployments.
- Designed RL-based agent training roadmap using GRPO policy optimization and custom RL Gym sandbox for training AI-driven workflow agents on simulated workspace interactions.
- Built real-time messaging with WebSocket-based communication and threaded channels in Go, supporting concurrent multi-user collaboration with low-latency message delivery.

ACHIEVEMENTS

Top 8 — OpenEnv AI Hackathon — Ranked top 8 out of 70,000+ developers in India's largest AI hackathon (Meta × Scaler SST × Hugging Face × PyTorch), building RL environments on Meta's OpenEnv. Evaluated by Meta & Hugging Face engineers; unlocked direct interview opportunity at Meta AI.

Intern of the Quarter — BLive — Recognized for technical excellence, ownership, and cross-team collaboration across fleet management and rider delivery platforms.

Winner — Scaler × OpenAI Hackathon — Won the national hackathon organized by Scaler School of Technology in collaboration with OpenAI.

TECHNICAL SKILLS

Languages: TypeScript · Python · Go · JavaScript · Dart · Java · C++ · SQL

Frontend: React · Next.js · Flutter · Tailwind CSS · React Native · GSAP · React Three Fiber

Backend: FastAPI · NestJS · Node.js · Go · PostgreSQL · MongoDB · Redis · Supabase · Firebase

Ai ML: LLM fine-tuning · Prompt Engineering · RAG · NLP · GRPO/RL · Multi-agent orchestration · Function-calling · OpenAI GPT · Anthropic Claude · Google Gemini · TRL · LangChain

Cloud Infra: Google Cloud Platform (GCP) · Docker · Git · Vercel · CI/CD · WebSockets · JWT Auth · REST APIs